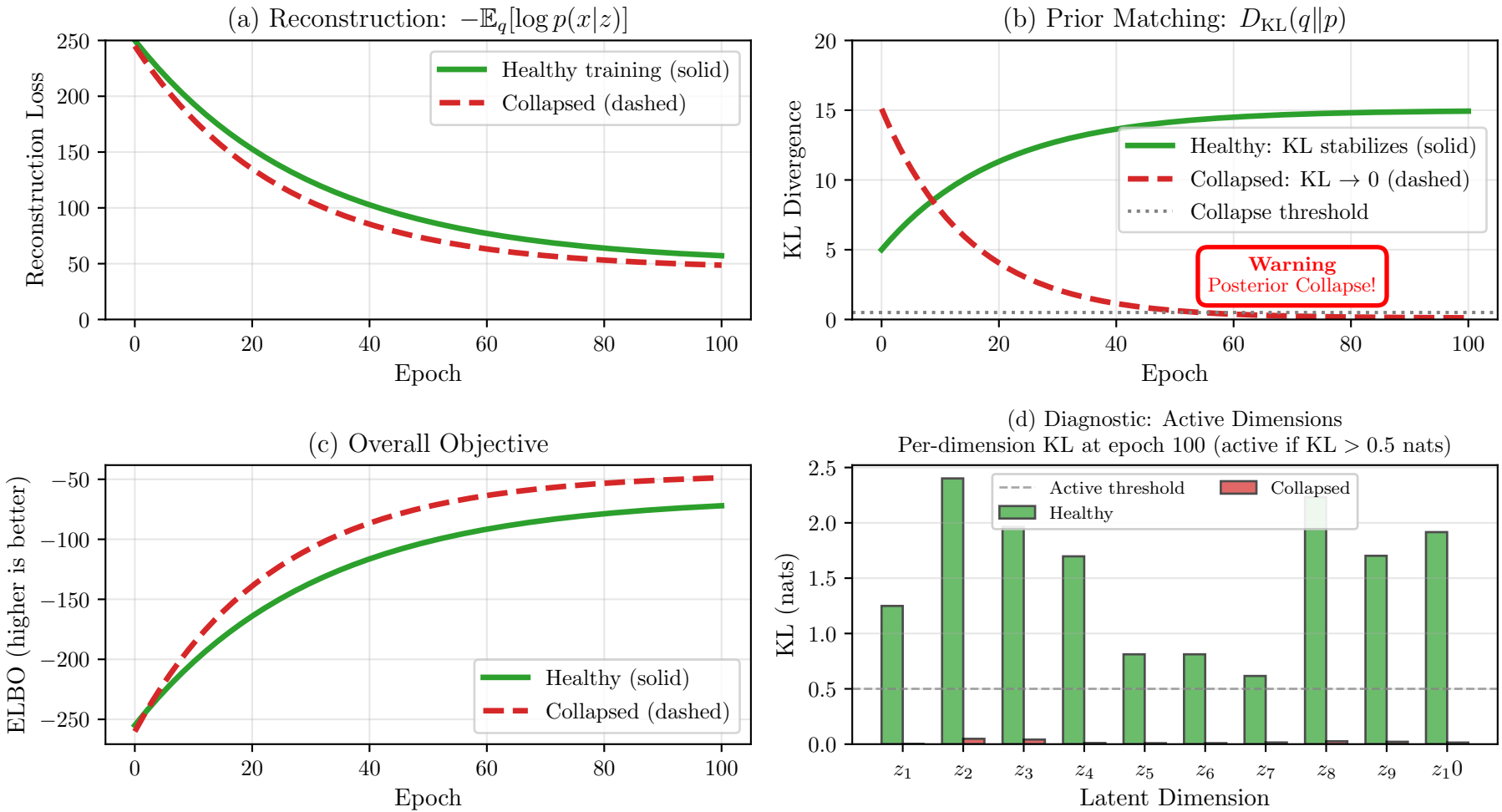
Posterior Collapse: Detection and Diagnosis
(a) Reconstruction: $-\mathbb{E}_q[\log p(x|z)]$
Reconstruction Loss
Epoch
Healthy training (solid)
Collapsed (dashed)
(b) Prior Matching: $D_{\mathrm{KL}}(q\|p)$
KL Divergence
Epoch
Healthy: KL stabilizes (solid)
Collapsed: KL → 0 (dashed)
Collapse threshold
Warning
Posterior Collapse!
(c) Overall Objective
ELBO (higher is better)
Epoch
Healthy (solid)
Collapsed (dashed)
(d) Diagnostic: Active Dimensions
Per-dimension KL at epoch 100 (active if KL > 0.5 nats)
KL (nats)
Latent Dimension
Active threshold
Collapsed
Healthy
z_1 z_2 z_3 z_4 z_5 z_6 z_7 z_8 z_9 z_{10}